Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/16/2004
PI Name:	Diddams, Scott A. Ph.D.		
Project Title:	Compact Femtosecond-Laser-Based Optical Synthesizers for Precision Measurements and Atomic Clocks in the Optical Domain		
Division Name:	Physical Sciences		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
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Atomic frequency standards that are being designed to fly aboard the International Space Station have projected fractional frequency uncertainties below 1e-16 with estimated instability as low as 3e-15 at 1 s of averaging. At the same time, emerging optical frequency standards based on laser-cooled atoms and ions promise even superior fractional frequency stability (<2e-16 at 1s of averaging) and uncertainty (1e-18). While it is valuable to operate these microwave and optical standards in isolated experiments, powerful tests of fundamental physical forces and symmetries can be designed around the comparison of standards operating at extremely different frequencies (i.e. microwave versus optical frequencies). Furthermore, there are several space-borne scientific and technical applications that could benefit from the extremely low fractional frequency instability provided by optical frequency standards if it could be transferred to the microwave domain. However, until very recently an appropriate high-performance synthesizer did not exist for

Task Description:	phase-coherently comparing and linking frequency standards in the optical and microwave domains. This research will attempt to fill this gap with science-enabling technological developments for high performance laser-cooled atomic time and frequency standards. The central objective is to develop femtosecond-laser-based technology into a compact, versatile, and robust optical frequency synthesizer. This ground-based study will involve the construction and testing of compact, low-power femtosecond laser sources, the stabilization of these lasers, and the determination of their relevant properties and fundamental limits. The outcome of the proposed research is anticipated to be a general-purpose device that can synthesize microwave frequencies from optical frequencies (and vice-versa) without degradation of the stability and accuracy of the reference input. This should prove valuable for fundamental physics experiments associated with the comparison of atomic clocks based on different transitions as well as the generation of very stable microwave signals. The central objective of this research is to develop femtosecond-laser-based technology into a compact and versatile optical frequency synthesizer. This ground-based study will center around the construction, testing, and optimization of a new femtosecond laser source that directly emits an octave-spanning spectrum. This will improve the robustness of these devices, making it especially suitable for applications in space. The outcome of the proposed research will be a general-purpose device that can synthesize microwave frequencies from optical frequencies (and vice-versa) without degradation of the stability and accuracy of the reference input. The femtosecond-laser-based optical frequency synthesizer is projected to support stabilities of 1e-16 at 1 s, and have a frequency uncertainty approaching 1e-18. This should prove valuable for fundamental physics experiments on earth and in space in at least three major areas Because they hold potential for significant stability and accuracy gains, there is good reason to believe that atomic clocks based on optical transitions will be the next generation of primary atomic frequency standards. The femtosecond-laser-based synthesizer proposed here is a crucial part of these future clocks. The potential for wide-spread positive impact can be extrapolated from the many ways in which precision time and frequency standards already play a fundamental role in the personal, commercial and scientific aspects of our society. Many civilian applications, and all branches of the military make extensive use of high precision atomic clocks. Applications of precision frequency standards range from the obvious¯time keeping and references for navigation¯to advanced future uses such as space-based atomic clocks for worldwide time/frequency distribution and navigation. High stability frequency references play critical roles in communication and navigation systems including, anti-jamming, friend or foe identification, spread spectrum communications, and increased security. These new frequency references coupled with the femtosecond-laser-based optical synthesizer will allow us to open the door to unexplored regions of science and space-time. Examples include searches for new physics such as graininess of space, and any time variation of atomic structure and fundamental constants. Some recent astronomical observations show evidence for a time variation of the fine structure constant, which if verified by earth or space-based measurements would have profound implications for our understanding of nature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Atomic clocks have been critical sources for advanced navigation and communications systems. This work will support the development of these technologies
Task Progress:	Funding for FY04 arrived in May of 2004. Nonetheless, we were able to complete all of our milestones for FY04 and made good progress on the milestones for FY05 Significantly we have: 1. Constructed and compared two octave-spanning femtosecond laser synthesizer (alternatively call femtosecond laser frequency combs, FLFC's). We have compared our two FLFC's against each other in addition to comparing them with two independently constructed FLFC's from France and China. The result of these comparisons was that when used for optical-to-optical frequency synthesis covering ~200 THz, the fractional uncertainty introduced by the FLFC (relative to a controlling optical frequency standard) is ~1e-19. This result is significant because it implies that the FLFC will not pose a limitation in the comparison of emerging optical frequency standards at or below the level of 1e-18 2. We have demonstrated that the elments (~1e6) of the FLFC can have linewidth significantly less than 1 Hz when referenced to a suitably narrow optical standard having similar or lesser linewidth. This result was achieved by optimization of the control servos of the FLFC and minimizing perturbations in the measurement apparatus. The significance of this result is that it will permit the generation of sub-Hz optical frequenciy references across the entire visible and near infrared portion of the spectrum. 3. We have identified amplitude to phase noise conversion as the primary source of excess noise in the photodetection of ultrastable optical pulse trains. We have mitigated this noise through improved laser design (i.e. quieter lasers), yet anticipate that active amplitude control will likely be necessary in the future. Nonetheless, in the current state, we have preliminary data that indicate we should be able to generate 1 or 10 GHz microwave electronic signals (directly from the photodetectors) having instability approaching 1e-15 at 1 s of averaging.
Bibliography Type:	Description: (Last Updated: 05/13/2021)
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